

REQ ID	TYPE	DESCRIPTION	PRIORITY	ACCEPTANCE CRITERIA	RATIONALE
FR-001	Functional	The system shall automatically queue uploaded student project zip files, execute pre-configured unit test suites, and generate an itemized rubric score breakdown.	High	Pass: Rubric breakdown is rendered with test results within 10 seconds of submission; Fail: File queue stalls or invalid rubric tally.	automates grading pipeline without manual evaluator effort.
FR-002	Functional	The system shall allow a student to upload a project submission as a .zip file through a web interface before the assignment deadline.	High	Pass: file is accepted and a submission timestamp is recorded when uploaded before deadline; Fail: system accepts a submission after the deadline, or rejects a valid .zip.	Without a defined upload path and deadline check, FR-001's auto grading has nothing to queue, and late submissions could be graded unfairly alongside on time ones.
FR-003	Functional	The system shall display the itemised rubric score breakdown and test results to the student who submitted the project	High	Pass: student can view their own rubric breakdown and test pass/fail results after grading completes; Fail: student sees no results, or can view another student's results.	FR-001 generates the rubric data on the backend only. A separate display requirement is needed because generating data and showing it to a user are two different system functions.
FR-004	Functional	The system shall allow a faculty evaluator to review and manually override the auto generated rubric score for any student submission.	Medium	Pass: faculty edits a score, the change is saved, and the student's displayed result reflects the new score; Fail: override is not saved, or applies to the wrong submission.	Automated scoring (FR-001) can misjudge edge cases , therefore faculty needs a way to correct this.
FR-005	Functional	The system shall detect when a submission's test suite fails to execute , (ex syntax errors) and flag the submission as "ungraded : execution error" instead of assigning a rubric score.	High	Pass: a submission that cannot compile/run is marked "execution error" and excluded from scoring; Fail: system assigns a rubric score to a submission that never ran, or crashes instead of flagging it.	FR-001 assumes the test suite runs successfully. Without this requirement, a broken submission would either crash the pipeline or silently receive an incorrect score.
NFR-001	Non-Functional (Security and Performance)	The system shall handle up to 100 concurrent project submissions without dropping queue items or exceeding 1.5 GB memory footprint.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load; Fail: system drops queue items, exceeds 1.5 GB memory, or fails security checks at 100 concurrent submissions.	A grading system used by an entire class receives many simultaneous submissions near the deadline. Without this requirement, the system could crash or slow down exactly when most students are submitting.
NFR-002	Non-Functional (Data Integrity)	The system shall persist all rubric scores and execution results such that no submitted grade is lost or altered without an audit record, even in the event of a server restart or crash during grading.	High	Pass: after a forced server restart mid-grading, all previously completed scores remain intact and retrievable; Fail: any score is lost, silently changed, or unrecoverable after a crash.	FR-001 and FR-004 both produce or modify scores. Without guaranteed persistence, a crash or override could silently corrupt a student's grade with no way to detect or reverse it.